

Network and Systems all Exercises

▼ Lecture 4+5

▼ Exercise 1

Exercise 1

- N=8 sensors each produce periodically 100 bits of measurements data every T=1mS.
- Data are to be sent across a shared communication line offering s= 1Mbps of serialization speed (bandwidth).
- Data from each sensor will be packetized with N (tbd) data points together with an overhead (header, error detection ...) of O=100 bits
- Determine the best value for N taking into account delay and bandwidth.
- For arbitrary (variable) N, determine a DNC arrival constraint for each sensor as well as the aggregate arrival curve for all sensors.

How many bits of in one packet per sensor:

$BW \text{ d } 1e6 :$

$nSensors \text{ d } 8 :$

$fs \text{ d } 1000 :$

$Overhead \text{ d } 100 :$

$solve(BW = (x \cdot C \cdot Overhead) \cdot nSensors \cdot fs, x)$

25.

(1.1.1)

$Data(t) \text{ d } 100 :$

$SentData(t) \text{ d } 125 \cdot L \cdot 100 :$

$solve(Data(t) = SentData(t), t)$

4

(1.1.2)

Number of meas per packet:

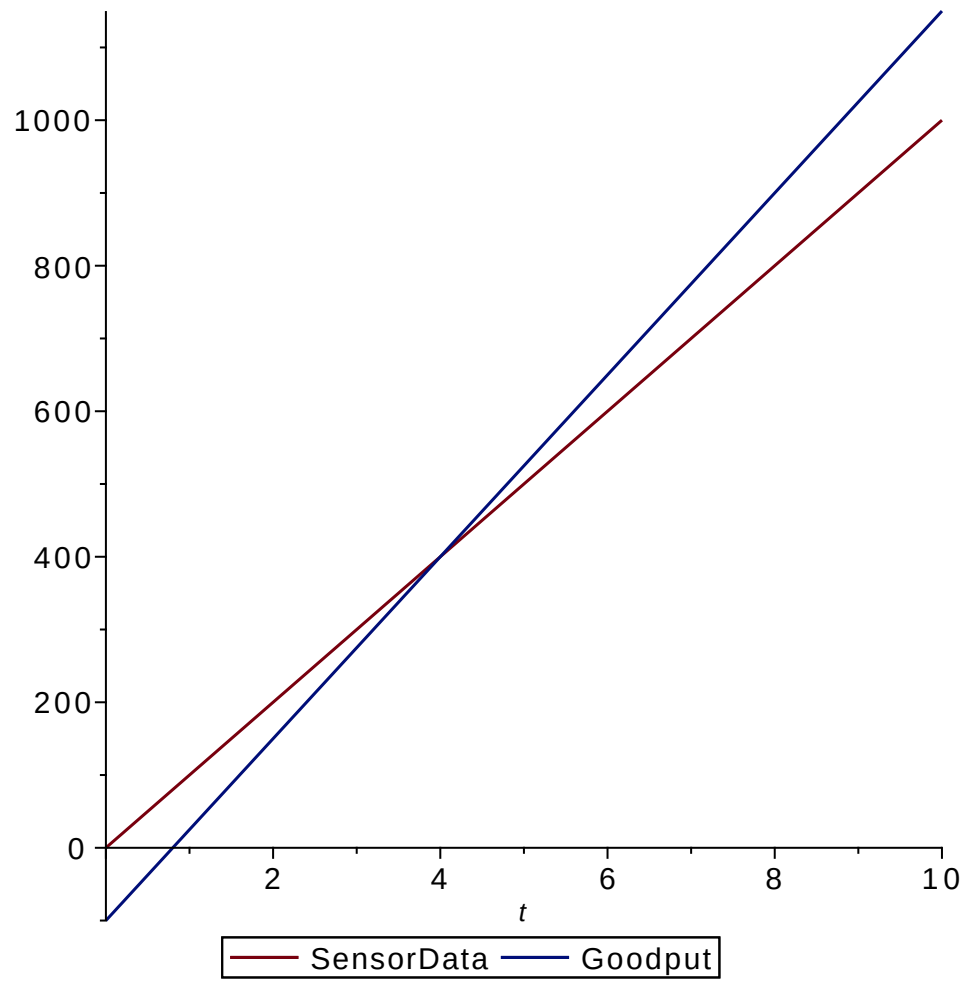
$solve\left(BW = (x \cdot 100 \cdot C \cdot Overhead) \cdot \left(\frac{fs}{x}\right), x\right)$

4.

(1.1.3)

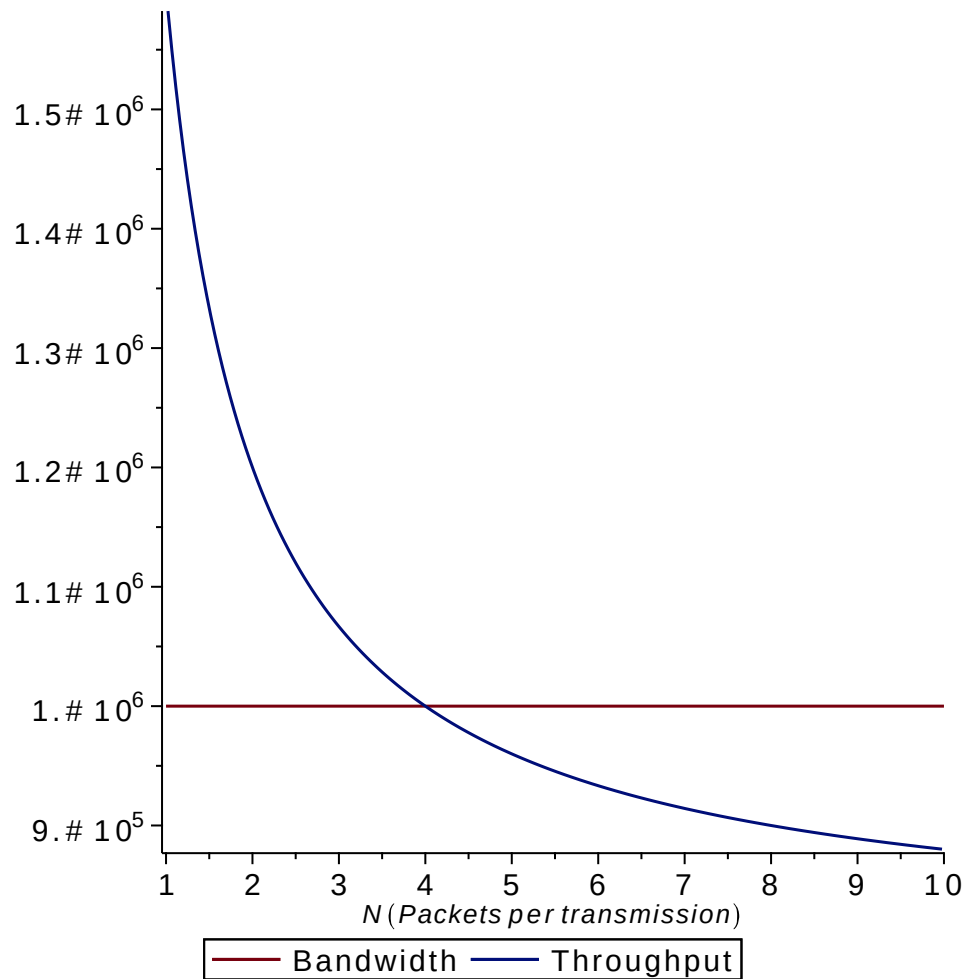
Less N --> overloaded, N = 4 --> saturated, N > 4 Spare capacity

$plot(\{Data(t), SentData(t)\}, t = 0 .. 10)$



Throughput curve cross BW 1Mbit @ N = 4

$plot\left(\left\{1e6, (x-100C\ Overhead)\frac{1000}{x}\right\}, x = 1..10\right)$



Aggregated packet arrivals N: 1 - 8
 Red = Service curve
 Blue = Received packets (Bursts)
 If blue > red --> Queue gets overloaded

for n **from** 1 **to** 8 **do**

$F_n \leftarrow \left\lfloor \frac{t}{n} \right\rfloor (n \cdot 1000 \cdot C \cdot 100) :$

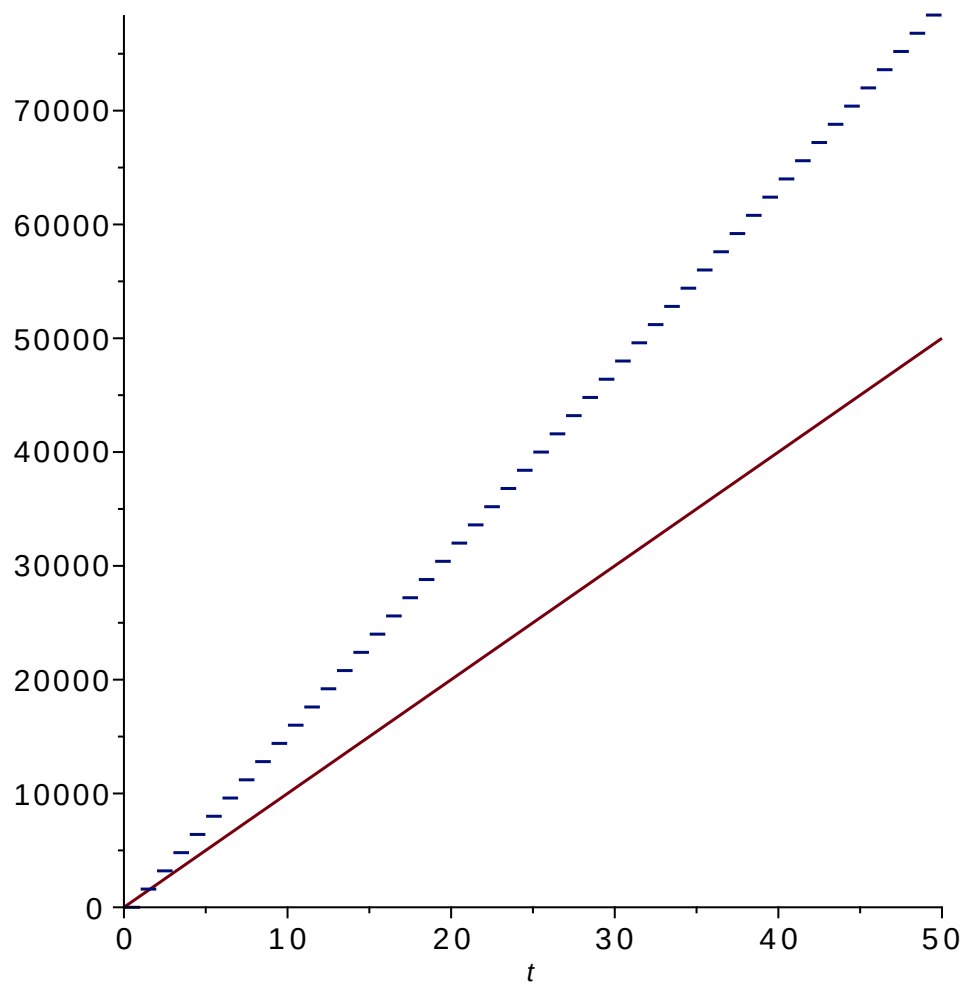
enddo:

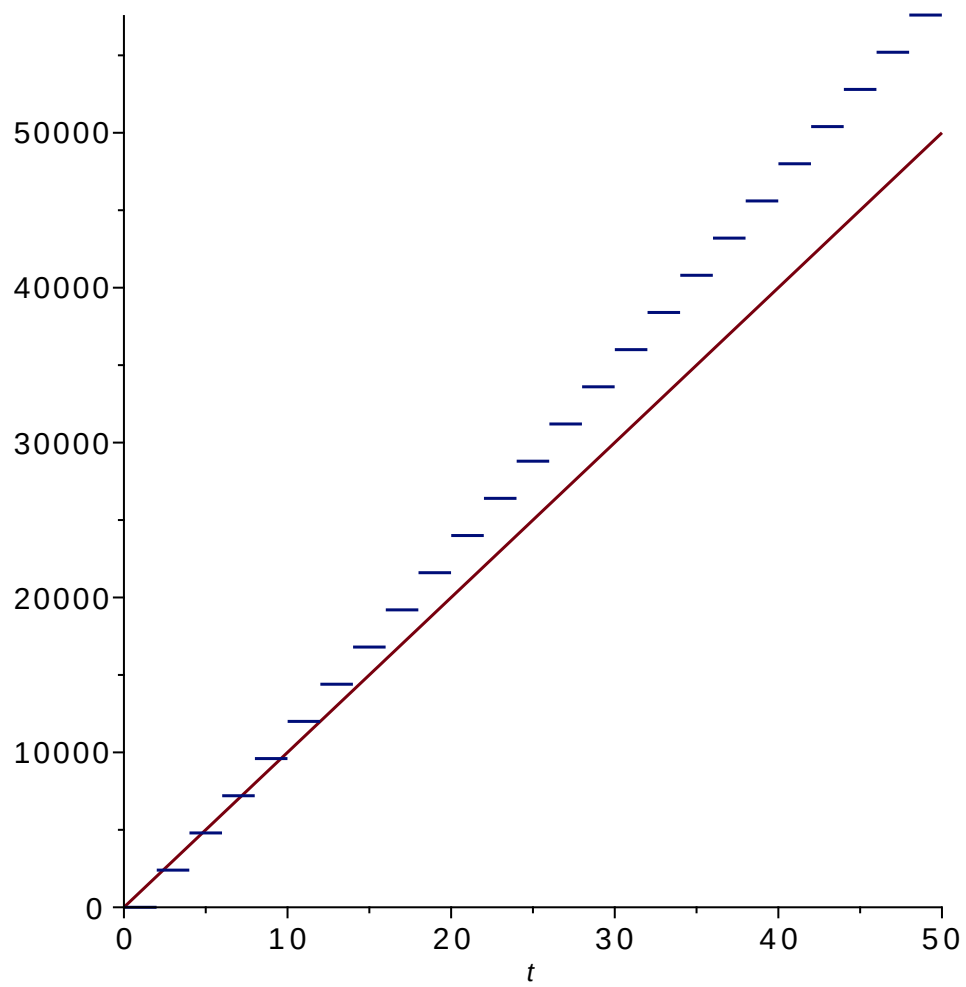
$\maxRate(t) \leftarrow \frac{BW}{1000} :$

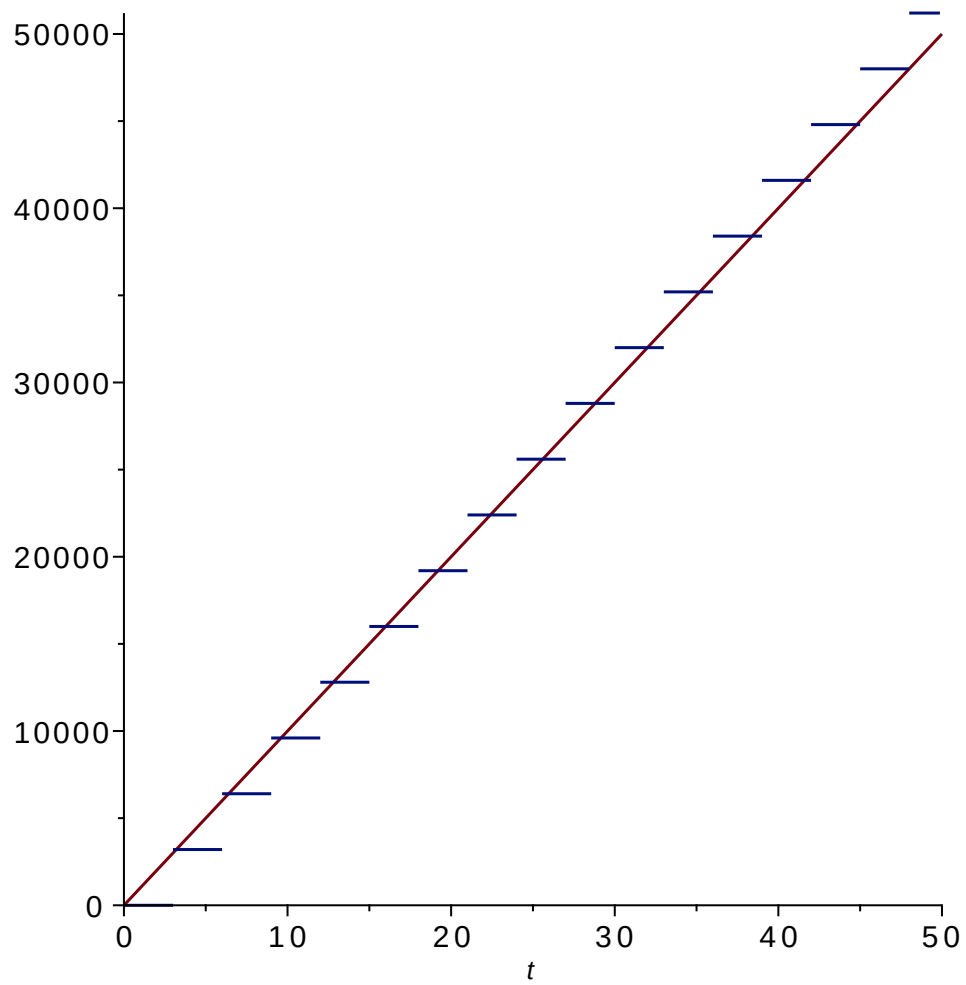
for n **from** 1 **to** 8 **do**

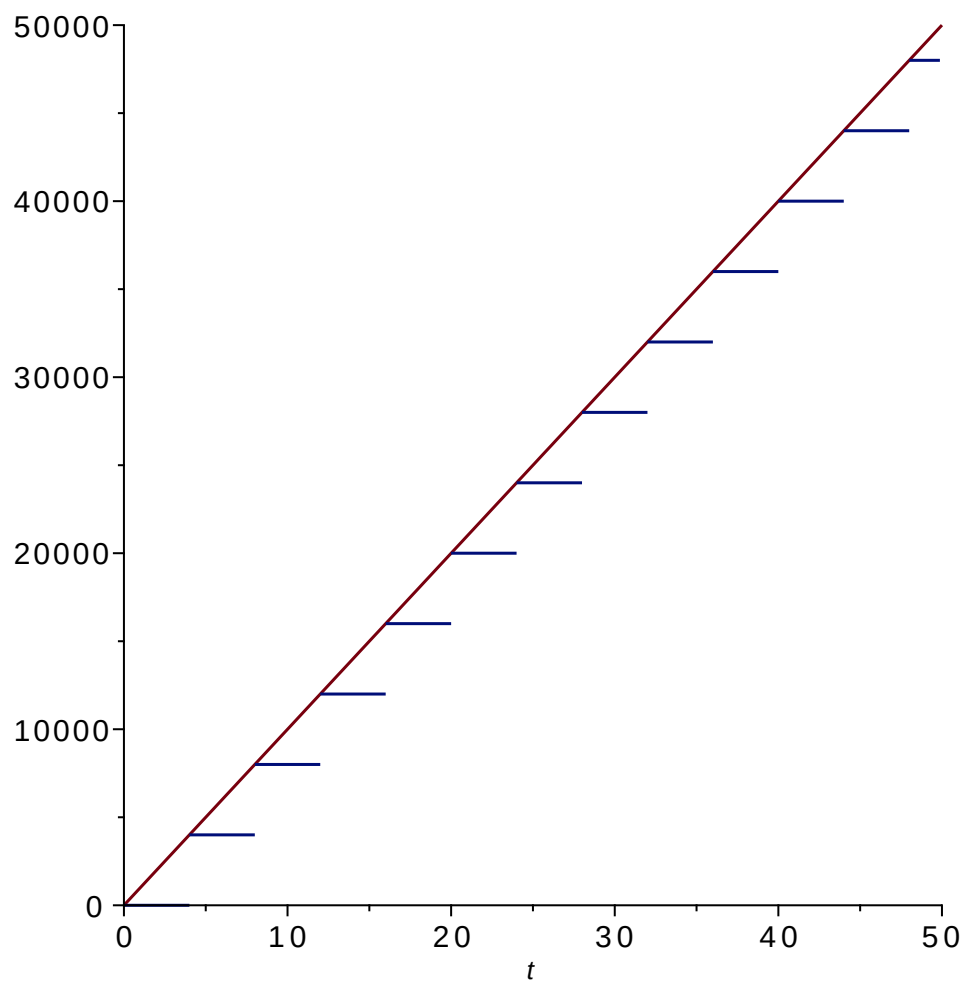
$plot(\{\maxRate(t), 8F_n\}, t = 0..50)$

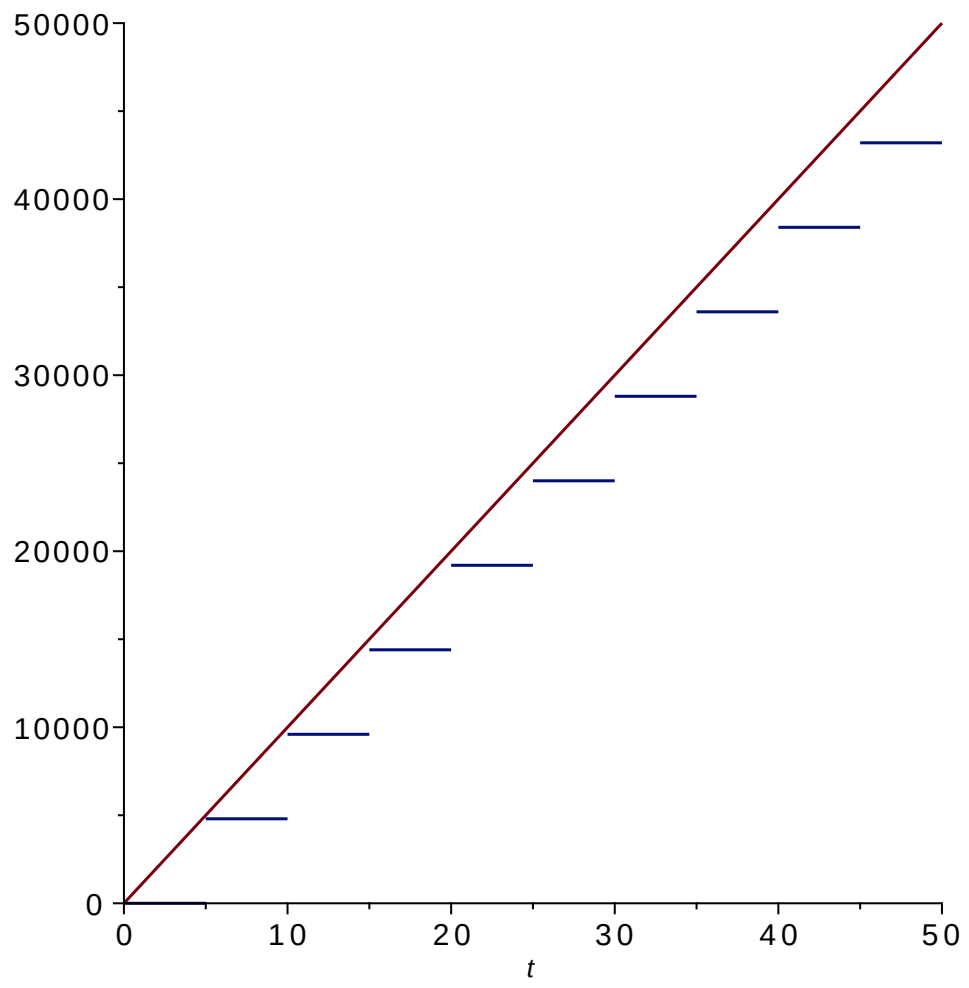
enddo

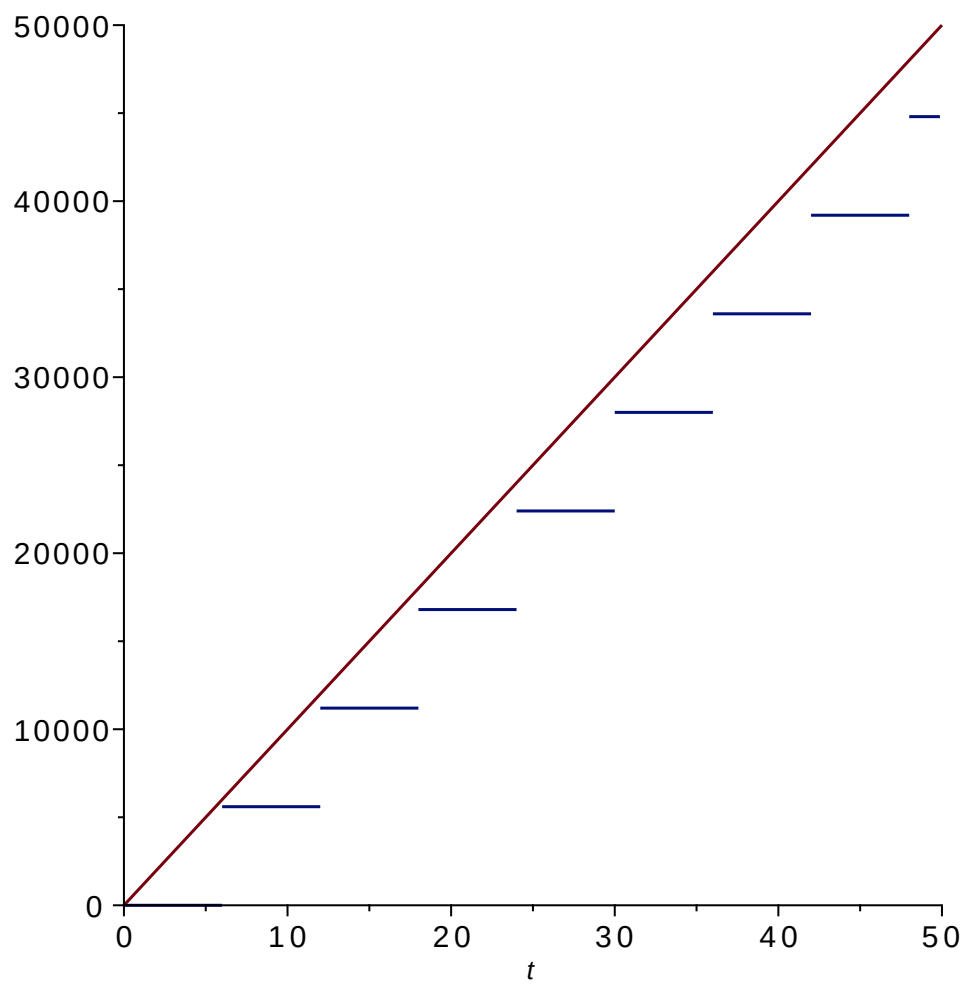


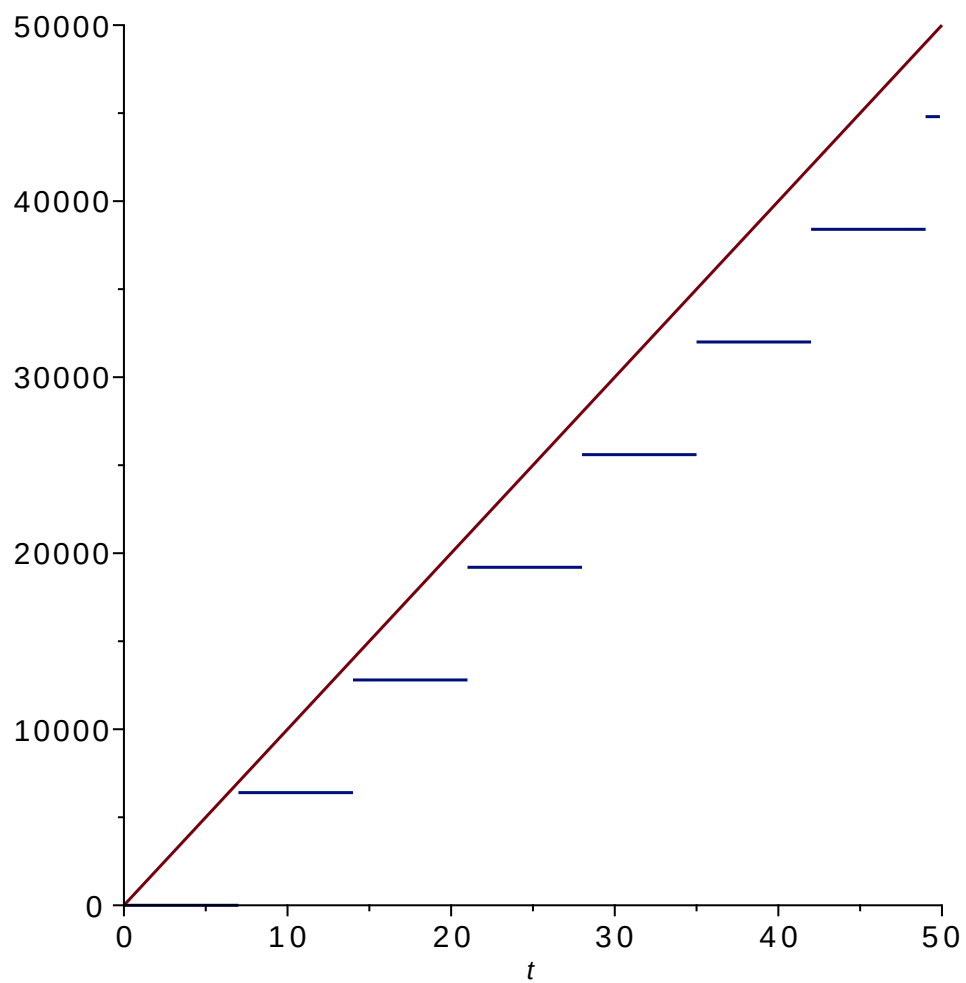


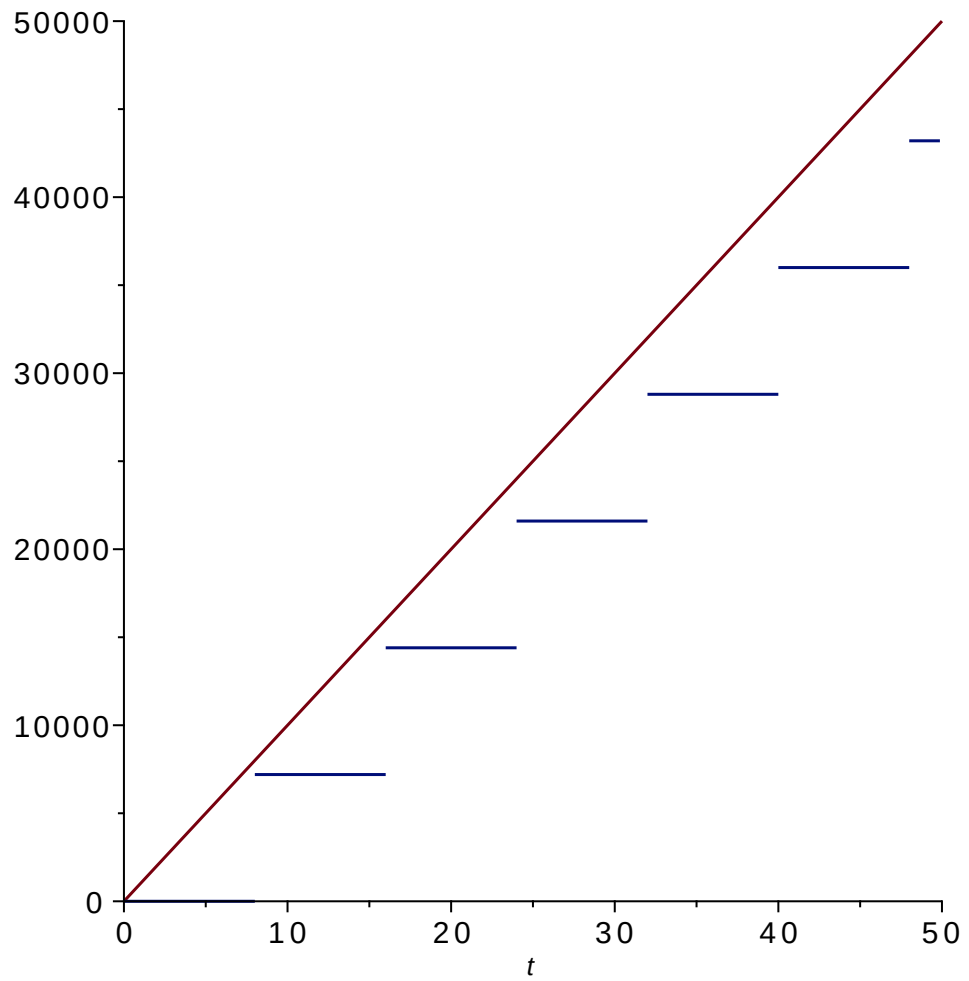






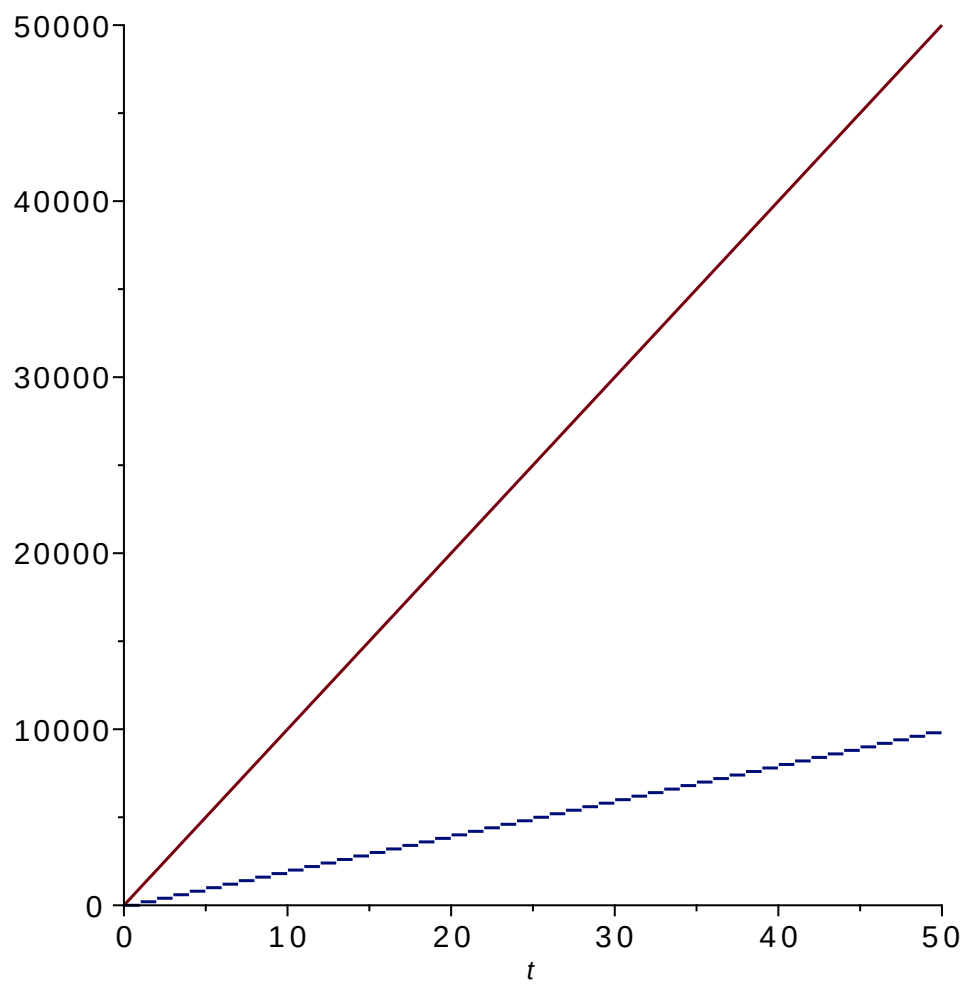


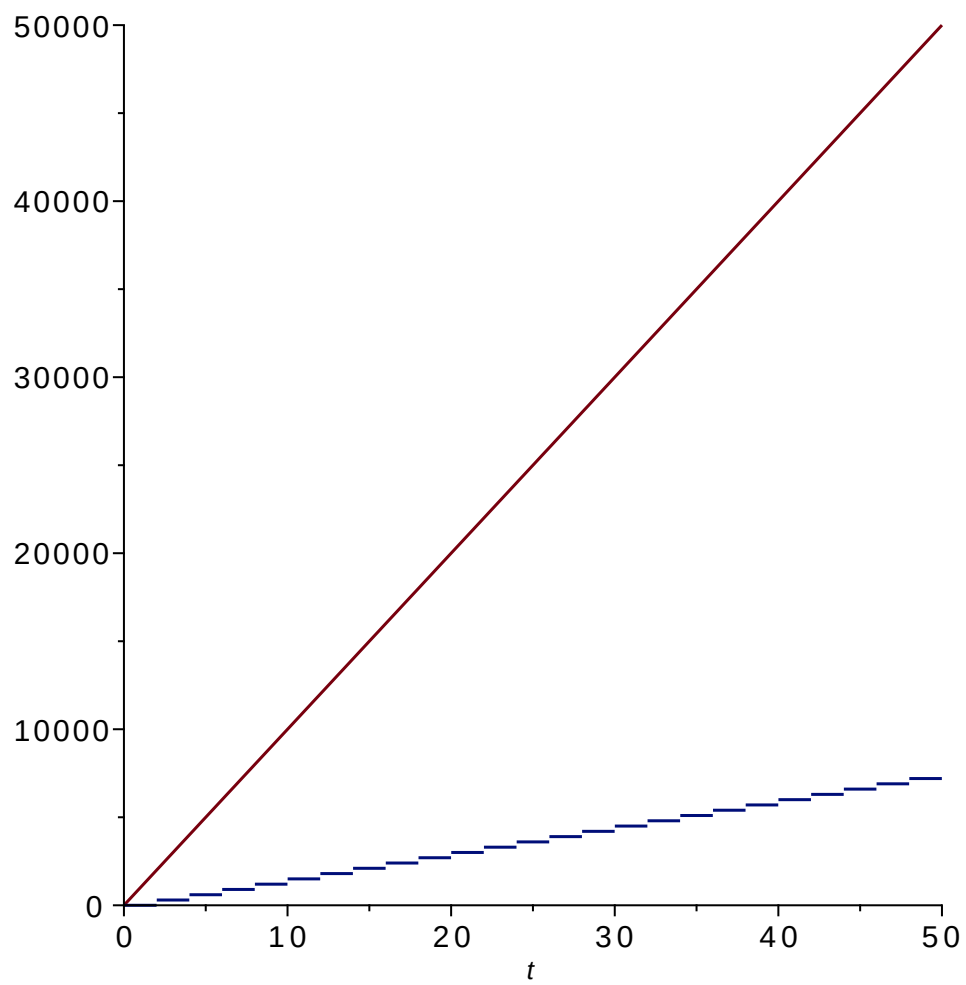


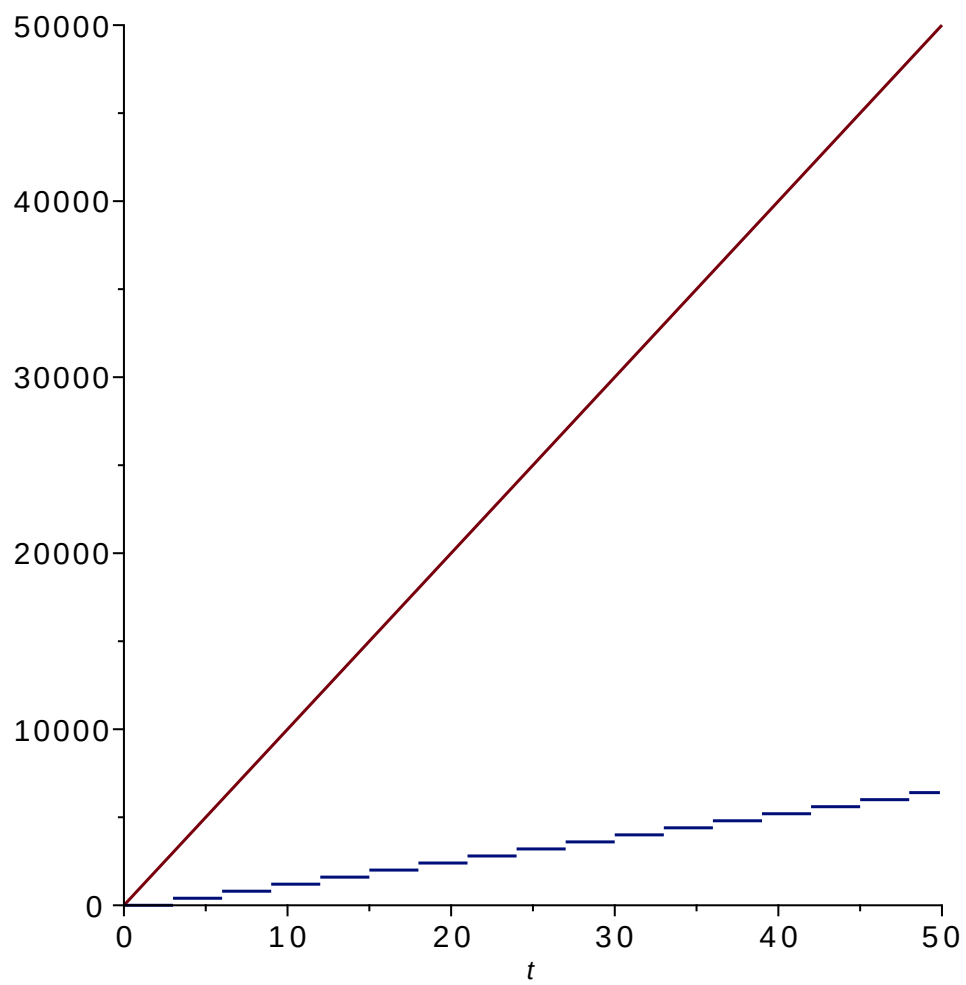


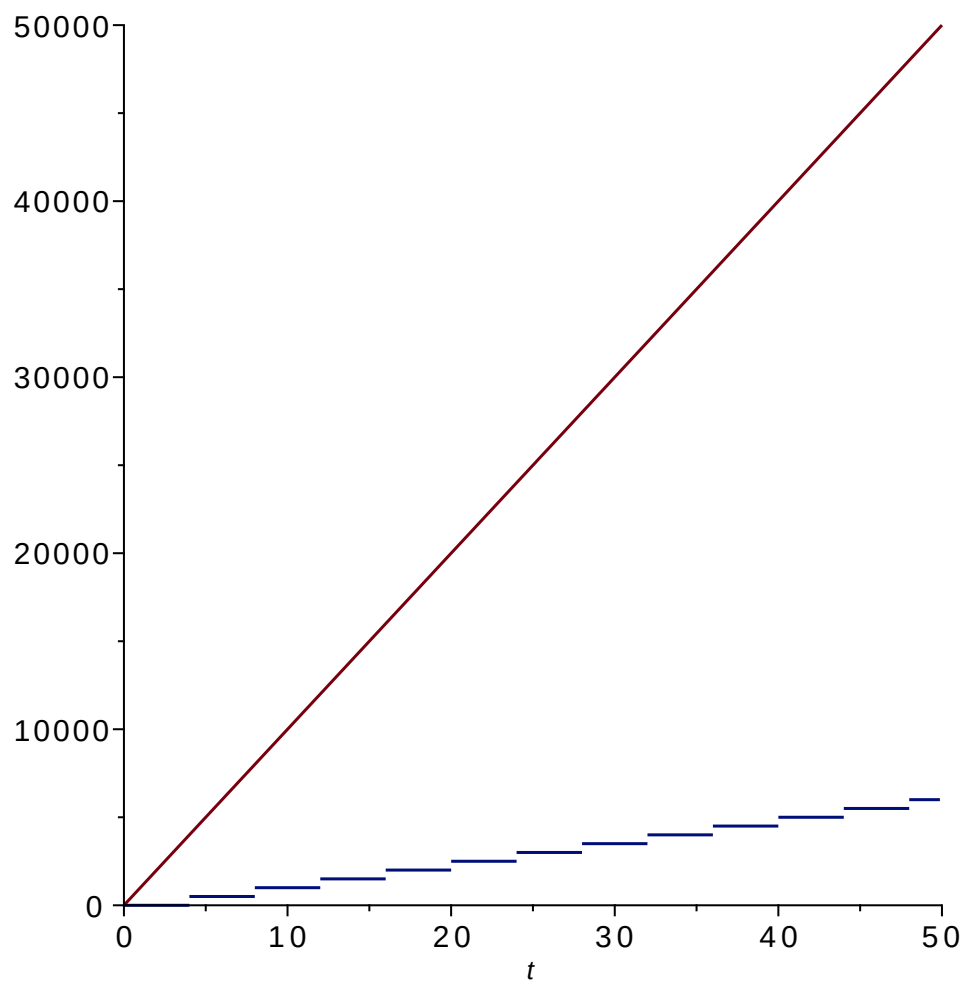
Plots of individual sensors:

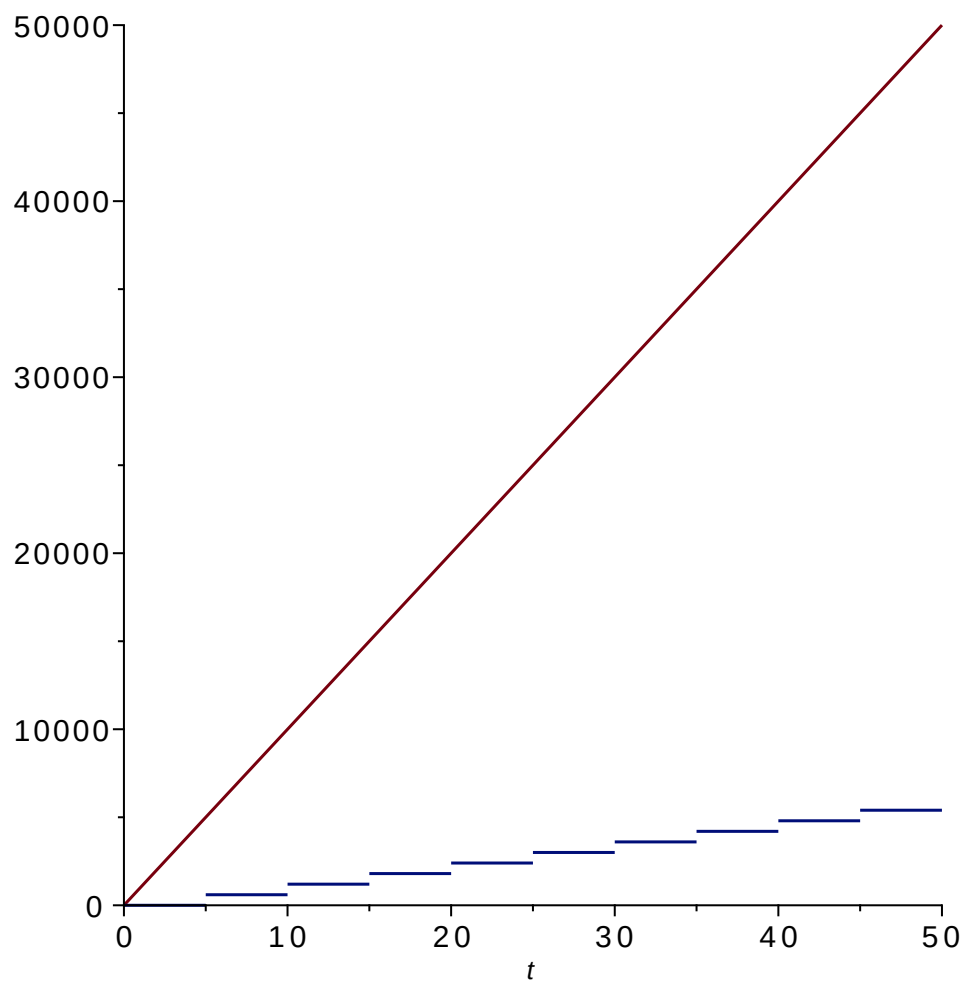
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for  $n$  from 1 to 8 do
   $plot(\{maxRate(t), F_n\}, t = 0..50)$ 
enddo
```

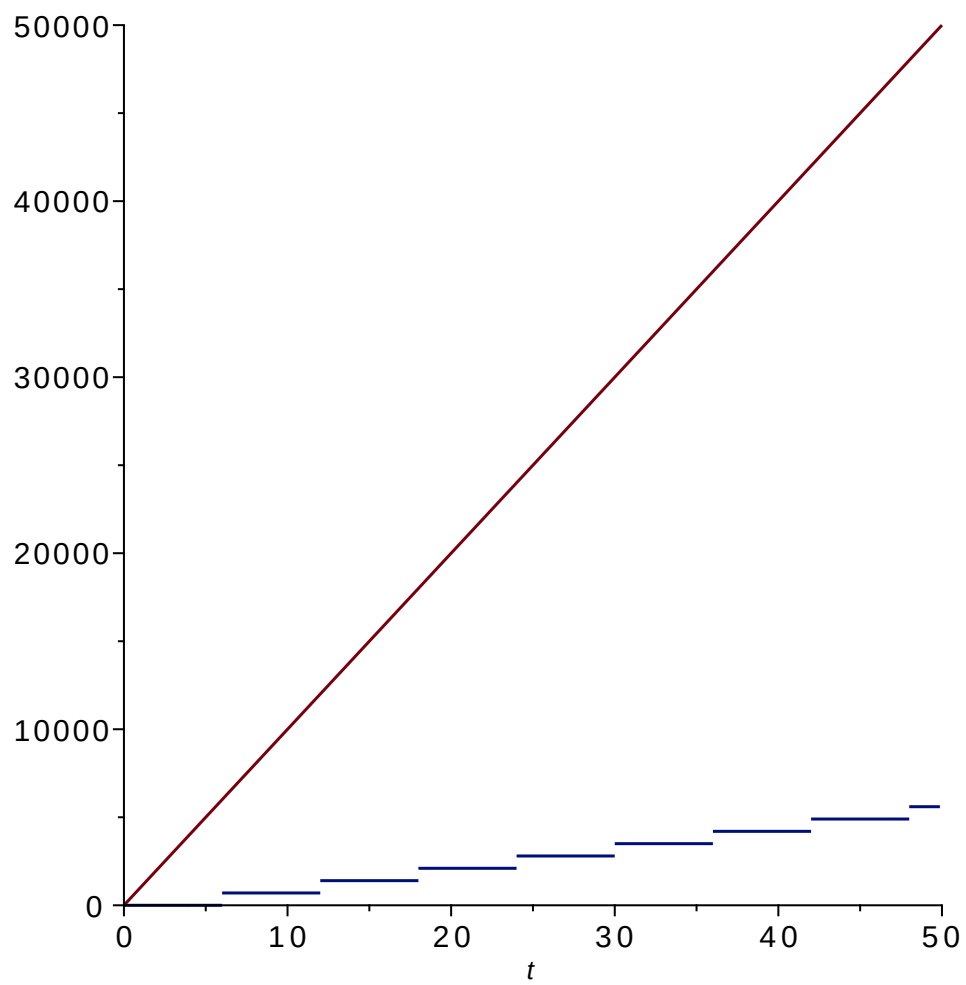


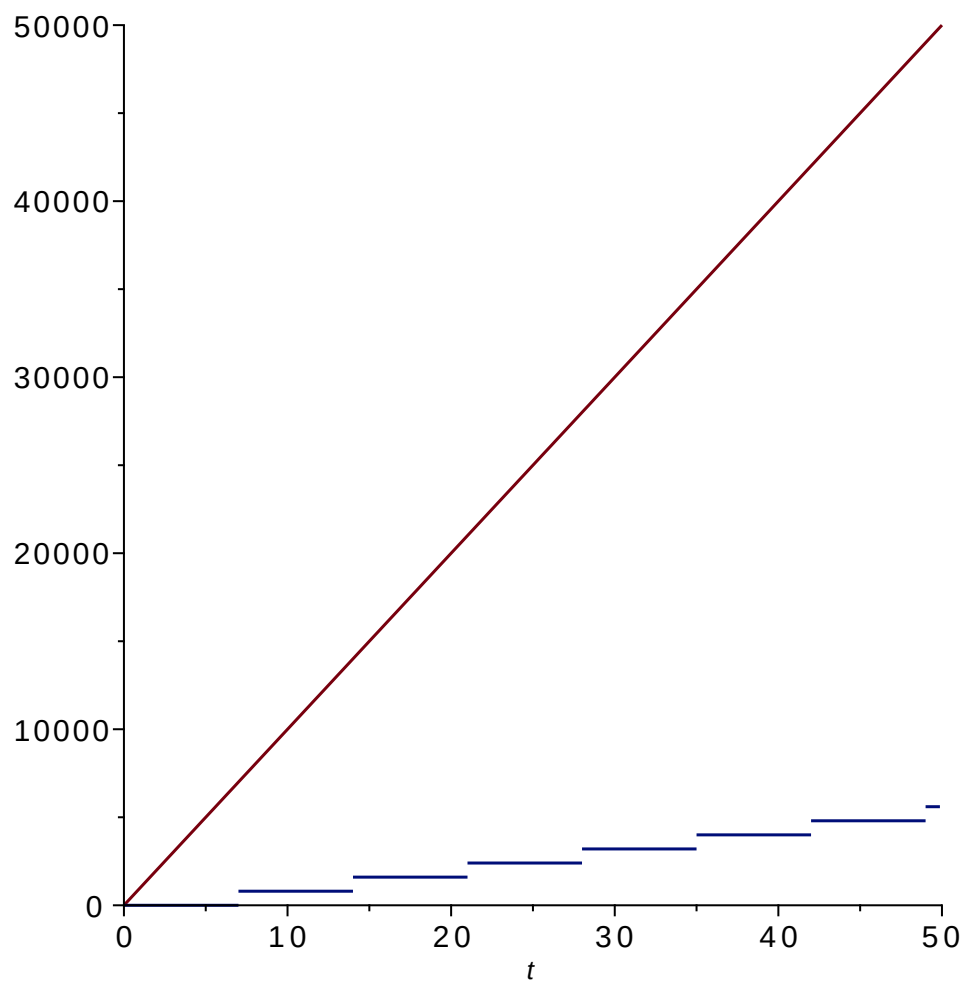


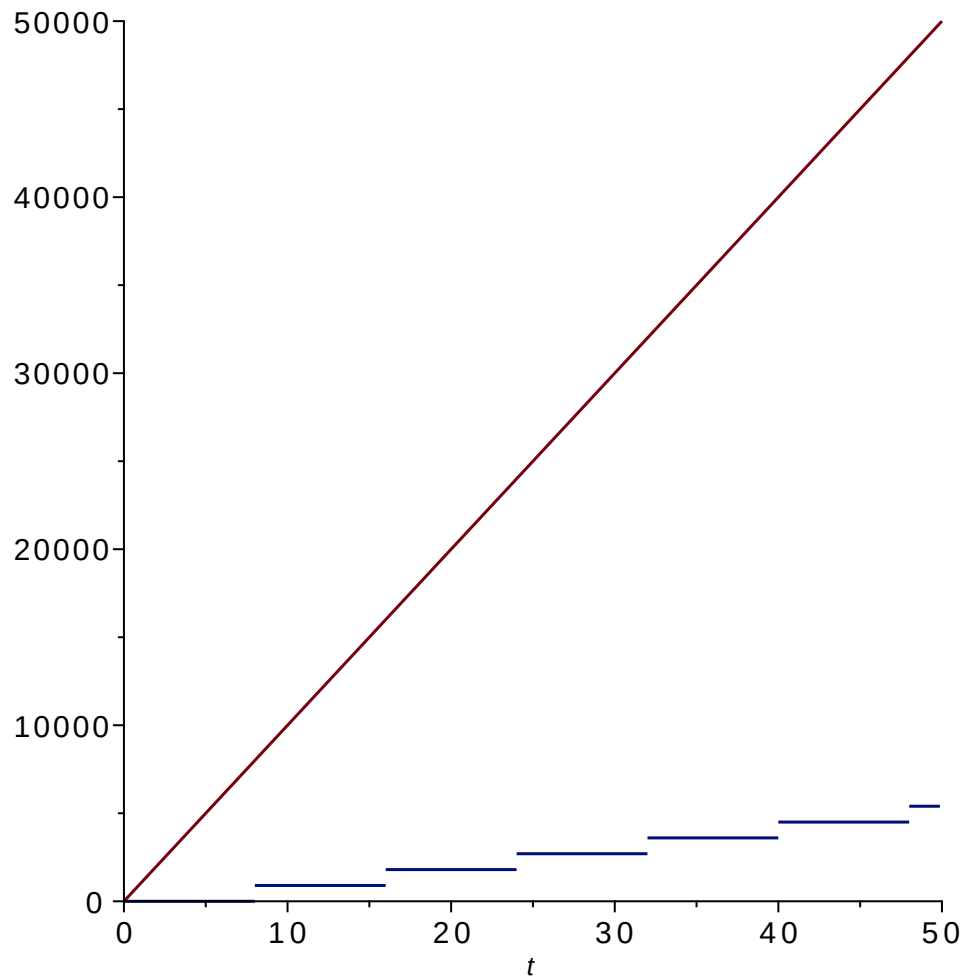












▼ Exercise 2

- Consider 3 nodes on a single CANbus segment with the following periods and priorities:
 - 0: {2mS, 000...}
 - 2: {3mS, 0002...}
 - 1: {5mS, 0001...}
- Assume packet lengths of 100 bits.
- Determine the smallest bitrate on the CANbus for the traffic to be executable.
- Make an RTC-model for the 3 nodes and CANbus with arbitrary bitrate. Include a receive buffer in the lowest priority node as well as a periodic receiver with arbitrary period.
- Determine maximum p2p delay for node 2 as well as minimum receive buffer (for not losing packets) for arbitrary receive periods and bitrates.

data_length d 100:

sensor_0_Ts d 2 10^3 :

sensor_1_Ts d 5 10^3 :

sensor_2_Ts d 3 10^3 :

$$\text{total_packets_per_second} = \frac{1}{\text{sensor_0_Ts}} + \frac{1}{\text{sensor_1_Ts}} + \frac{1}{\text{sensor_2_Ts}} =$$

$\frac{3100}{3} \xrightarrow{\text{at 5 digits}} 1033.3$

$$\text{total_bit_per_second} = \text{total_packets_per_second} \times \text{data_length} = 103.33 \times 10^3$$